

Deployment & Configuration Guide

FX Trading Tracker Platform | On-Premise Setup, Environment Variables & Network Requirements

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1. Prerequisites

Before deploying, ensure the following are available in the target environment:

Requirement	Version	Notes
Python	3.11.15	Runtime for FastAPI backend
Node.js	20.20.2	For building React frontend
MongoDB	7.0.31	Default database (or Couchbase Enterprise)
Couchbase Enterprise	7.0+	On-premise database (optional, SDK 4.6.0)
S3-Compatible Storage	Any	Huawei OBS, MinIO, or AWS S3 for file storage
Supervisor	Any	Process manager for frontend and backend services
HTTPS Certificate	TLS 1.2+	For production HTTPS termination

2. Environment Configuration

2.1 Backend (.env)

Create /app/backend/.env with the following variables. All values are required unless marked optional.

Variable	Required	Description
MONGO_URL	If DB_TYPE=mongodb	MongoDB connection string
DB_NAME	If DB_TYPE=mongodb	MongoDB database name
DB_TYPE	Yes	mongodb or couchbase
CB_CONNECTION_STRING	If DB_TYPE=couchbase	Couchbase cluster connection string
CB_USERNAME	If DB_TYPE=couchbase	Couchbase username
CB_PASSWORD	If DB_TYPE=couchbase	Couchbase password
CB_BUCKET_NAME	If DB_TYPE=couchbase	Couchbase bucket name
JWT_SECRET	Yes	Random secret for JWT signing (min 32 chars)
STORAGE_TYPE	Yes	emergent or s3
S3_ENDPOINT_URL	If STORAGE_TYPE=s3	S3-compatible endpoint
S3_ACCESS_KEY_ID	If STORAGE_TYPE=s3	S3 access key
S3_SECRET_ACCESS_KEY	If STORAGE_TYPE=s3	S3 secret key
S3_BUCKET_NAME	If STORAGE_TYPE=s3	S3 bucket name
S3_REGION	If STORAGE_TYPE=s3	S3 region

2.2 Frontend (.env)

Set REACT_APP_BACKEND_URL to the production base URL (e.g., https://fx.company.com).

3. Switching Database Backend

3.1 MongoDB (Default)

Set DB_TYPE=mongodb in backend/.env. Ensure MONGO_URL and DB_NAME are configured. The application auto-seeds default users and reference data on first startup.

3.2 Couchbase Enterprise

Set `DB_TYPE=couchbase`. Configure `CB_CONNECTION_STRING`, `CB_USERNAME`, `CB_PASSWORD`, and `CB_BUCKET_NAME`. On first startup, the application automatically: (1) Creates 4 scopes (identity, trading, reference, audit), (2) Creates 10 collections, (3) Creates primary and secondary indexes, (4) Seeds default users and reference data. Ensure the Couchbase user has Data Reader/Writer and Query/Index permissions on the bucket.

4. Switching Storage Backend

4.1 S3-Compatible (Huawei OBS / MinIO)

Set `STORAGE_TYPE=s3` and configure `S3_ENDPOINT_URL`, `S3_ACCESS_KEY_ID`, `S3_SECRET_ACCESS_KEY`, `S3_BUCKET_NAME`, `S3_REGION`. The bucket must exist before starting the application. The S3 client uses s3v4 signature and path-style addressing for maximum compatibility.

5. Default User Accounts

The following accounts are auto-created on first startup if no users of that role exist:

Role	Email	Password
Admin	admin@fxtracker.com	Admin@123
Trader	trader@fxtracker.com	Trader@123
Treasury	treasury@fxtracker.com	Treasury@123

6. Network Requirements

Port	Service	Direction
3000	Frontend (React)	Inbound (via reverse proxy)
8001	Backend (FastAPI)	Inbound (via /api prefix routing)
27017	MongoDB	Backend to DB (if MongoDB)
11207	Couchbase KV (TLS)	Backend to DB (if Couchbase)
18091	Couchbase Mgmt (TLS)	Backend to DB (if Couchbase)
18093	Couchbase Query (TLS)	Backend to DB (if Couchbase)
9000	S3/MinIO	Backend to Storage (if S3)